Induced Subgraphs in Ramsey Graphs

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Examination Number: XXXXXXXX
Degree Course: MSc in Mathematical Sciences
Submission: Trinity Term 2026

Submitted in partial fulfilment of the requirements for the degree of Master in Mathematical Sciences (OMMS).

Word count:

Abstract

In this dissertation, we investigate the "random-like" features of Ramsey graphs, with an emphasis on the existence of induced subgraphs of many different sizes.

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1 Introduction

Definition 1.1. Let G be a graph. A set $S \subseteq V(G)$ is *homogeneous* if it induces a clique or an independent set in G.

Theorem 1.1 (Erdős–Szemerédi [3])**.** *Let G be a graph on n vertices with less than dn^2 edges. Then G contains a homogeneous set of size greater than $\frac{c}{d\log(1/d)}\log n$.*

Proof. Note that there exists $m \geq n/2$ vertices $v_1, v_2, \ldots, v_m$ with $\deg(v_i) \leq 2dn$. Let H be the graph induced by these vertices. Rearranging

Let k be the size of the largest independent set in H □

Corollary 1.2. *For any C-Ramsey graph G, there exists $\delta = \delta(C) \in (0, 1/2)$ such that the edge density of G lies in the interval $[\delta, 1-\delta]$.*

Proof. Apply Theorem 1.1 to G and $\overline{G}$. □

2 Subgraphs with Many Degrees

In this chapter, we present the following result of Bukh and Sudakov.

Theorem 2.1 (Bukh–Sudakov [1])**.** *Every Ramsey graph on n vertices induces a subgraph on $\Omega(n)$ vertices with $\Omega(\sqrt{n})$ distinct degrees.*

This was originally conjectured by Erdős, Faudree, and Sós [4] in light of the observation that the random graph $G_{n,1/2}$ exhibits such a property almost surely, but the $\Omega(\sqrt{n})$ bound is in fact not optimal for both Ramsey graphs and $G_{n,1/2}$. In particular, an upper bound of $O(n^{2/3})$ is noted in [1], which was later proved to be the correct order for Ramsey graphs by Jenssen, Keevash, Long, and Yepremyan [5] via a continuous relaxation approach. The corresponding order for $G_{n,1/2}$ was also matched in an unpublished work of Conlon, Morris, Samotij, and Saxton [2], further reinforcing the case that Ramsey graphs resemble random graphs in various respects. We will, however, focus on the original result of Bukh and Sudakov, as it introduces a key characteristic of Ramsey graphs that will be insightful in the following chapter.

Definition 2.1. An n-vertex graph G is (c, δ)-*diverse* if for $x \in V(G)$, we have $|N(x)\Delta N(y)| < cn$ for at most n^δ vertices $y \in V(G)$.

The notion of diversity captures the idea that the neighbourhoods of most pairs of vertices differ significantly. In particular, the usefulness of diversity comes from bounding the number of vertices that shares the same degree into some random vertex set U, as most pairs of vertices would have fairly uncorrelated neighbourhoods into U.

The proof of Theorem 2.1 proceeds by showing that every Ramsey graph contains a diverse subgraph of linear size, followed by a probabilistic argument that diversity furnishes many distinct degrees.

Lemma 2.2. *For $\delta \in (0, 1)$, there exists $c = c(C, \delta) > 0$ such that every C-Ramsey graph G on n vertices contains a (c, δ)-diverse subgraph on $\Omega(n)$ vertices.*

Proof. Suppose not. Let $K = \lceil \gamma C \log(C + 1) \rceil$ for some sufficiently large constant $\gamma = \gamma(\delta)$, $m = 8^{-K}n$, and $c = 1/K$. We inductively construct a sequence of disjoint vertex sets $S_1, S_2, \ldots, S_K$ of size m^δ, such that for all S_i one of the following conditions holds:

(i) $d(S_i, S_j) < 8c$ for $j > i$,

(ii) $d(S_i, S_j) > 1 - 8c$ for $j > i$.

Alongside, we also construct a sequence of induced subgraphs $G = G_0 \subset G_1 \subset \cdots \subset G_K$, such that $|G_i| \geq |G_{i-1}|/8$ and $S_i \subseteq V(G_{i-1})$ for all i.

Suppose $G_1, \ldots, G_{i-1}$ and $S_1, \ldots, S_{i-1}$ have been constructed. By induction, $|G_{i-1}| \geq 8^{1-i}|G| \geq m$. Note that G_{i-1} is not (c, δ)-diverse by assumption, so there exist vertex $x \in V(G_{i-1})$ and set $S_i \subseteq G_{i-1}$ of size m^δ such that $|N_{G_{i-1}}(x)\Delta N_{G_{i-1}}(y)| < c|G_{i-1}|$

for $y \in S_i$. Let $T_i = V(G_{i-1}) \setminus S_i$. Since $m^\delta \le m/2$ for large enough n, we have $|T_i| \ge |G_{i-1}|/2$.

Suppose $|\overline{N}_{T_i}(x)| \ge |T_i|/2$. Then for $y \in S_i$,

$$d(y, \overline{N}_{T_i}(x)) = \frac{|N_{T_i}(y) \cap \overline{N}_{T_i}(x)|}{|N_{T_i}(x)|} < \frac{c|G_{i-1}|}{|T_i|/2} \le 4c.$$

Partition $\overline{N}_{T_i}(x)$ into $V_i \sqcup W_i$, where $V_i = \{v \in \overline{N}_{T_i}(x) : d(v, S_i) < 8c\}$. Notice

$$8c|W_i| \le \sum_{v \in W_i} d(v, S_i) \le \frac{e(\overline{N}_{T_i}(x), S_i)}{|S_i|} = \frac{|\overline{N}_{T_i}(x)|}{|S_i|} \sum_{v \in S_i} d(y, \overline{N}_{T_i}(x)) \le 4c|\overline{N}_{T_i}(x)|,$$

so $|W_i| \le |\overline{N}_{T_i}(x)|/2$. But then $|V_i| \ge |\overline{N}_{T_i}(x)|/2 \ge |T_i|/4 \ge |G_{i-1}|/8$. Set $G_i = G[V_i]$. Since $d(S_i, U) < 8c$ for all $U \subseteq V_i$, condition (i) holds for S_i. In the case that $|N_{T_i}(x)| < |T_i|/2$, the same argument applies by setting $V_i = \{v \in \overline{N}_{T_i}(x) : d(v, S_i) > 1 - 8c\}$, which yields $|G_i| \ge |G_{i-1}|/8$ and S_i satisfying condition (ii). This completes the induction.

By working on $\overline{G}$ instead of G if necessary, we may assume that $r \ge K/2$ of the sets $S_1, \ldots, S_K$ satisfy condition (i), say $S_{i_1}, \ldots, S_{i_r}$. Put $S = \bigsqcup_{k=1}^r S_{i_k}$. Since each S_{i_k} has size m^δ and $c = 1/K$,

$$d(S) \le \frac{\binom{m^\delta}{2}}{\binom{rm^\delta}{2}} + \max_{(k,l)} d(S_{i_k}, S_{i_l}) \le \frac{1}{r} + 8c \le \frac{10}{K}.$$

Thus Theorem 1.1 yields a homogeneous set A in S with $|A| \ge \frac{\nu K}{10 \log(K/10)} \log |S|$, for some $\nu = \nu(C)$. Note that $|A|$ is also a homogeneous set in G. But then

$$\log |S| \ge \frac{1}{\delta} \log m = \frac{1}{\delta} \log n - \frac{K \log 8}{\delta}.$$

Pick $\gamma > \nu K/10\delta \log(K/10)$ and we have $|A| > C \log n$ for large enough n, contradiction. □

Lemma 2.3. *Let $\delta \in (0, 1/2)$ and let G be a (c, δ)-diverse graph on n vertices for large enough n. Then for $n/4 \le m \le 3n/4$, there is a subgraph on m vertices containing vertices of $\sqrt{cn}/320$ distinct degrees.*

Proof. Fix m as above. Let U be chosen uniformly at random among all subsets of $V(G)$ of size m. Let $H = G[U]$. It suffices to show that with positive probability, H contains $\Omega(\sqrt{n})$ vertices of distinct degrees.

For distinct $x, y \in V(G)$, define $\mathbb{1}_{xy}$ to be the indicator for the event that $x, y \in V(H)$ and $\deg_H(x) \ne \deg_H(y)$. Then $\mathbb{E}[\mathbb{1}_{xy}] \le \mathbb{P}[\deg_U(x) = \deg_U(y)]$. We may write

$$\mathbb{P}[\deg_U(x) = \deg_U(y)] = \mathbb{P}\left[|(N_G(x) \setminus N_G(y)) \cap U| = |(N_G(y) \setminus N_G(x)) \cap U|\right].$$

Put $D = N_G(x)\Delta N_G(y)$, and $k = |D \cap U|$. Suppose $|D| \geq cn$. Fix an ordering on D and let X_i be the number of the first i vertices in $D \cap U$. Then $X_0, X_1, \ldots, X_{|D|} = k$ is a martingale with $|X_{i+1} - X_i| \leq 1$ for all i. Since $\frac{|D|}{4} \leq \mathbb{E}[k] = \frac{m|D|}{n} \leq \frac{3|D|}{4}$, the Hoeffding-Azuma inequality yields

$$\mathbb{P}\left[k < \frac{|D|}{8} \text{ or } k > \frac{7|D|}{8}\right] \leq \mathbb{P}\left[|k - \mathbb{E}[k]| > \frac{|D|}{8}\right] \leq e^{-\Theta(|D|)} = e^{-\Theta(n)} = o(n^{-1/2}).$$

But then

$$\mathbb{P}\left[\deg_U(x) = \deg_U(y) \,\middle|\, \frac{|D|}{8} \leq k \leq \frac{7|D|}{8}\right] = \frac{\binom{|N_G(x)\setminus N_G(y)|}{k/2}\binom{|N_G(y)\setminus N_G(x)|}{k/2}}{\binom{|D|}{k}}.$$

It now follows from Stirling's formula and $k(|D| - k) \geq \frac{7}{64}|D|^2$ that the above probabilitydoes not exceed

$$\frac{\binom{|D|/2}{k/2}^2}{\binom{|D|}{k}} = (1 + o(1))\sqrt{\frac{2|D|}{\pi k(|D| - k)}} \leq \frac{3}{\sqrt{|D|}} \leq \frac{3}{\sqrt{cn}}.$$

Thus if $|N_G(x)\Delta N_G(y)| \geq cn$ then $\mathbb{E}[\mathbb{1}_{xy}] \leq 3/\sqrt{cn} + o(n^{-1/2}) \leq 4/\sqrt{cn}$.

Let $I = \sum_{x,y \in V(G)} \mathbb{1}_{xy}$. By definition of a (c, δ)-diverse graph, we have $\mathbb{E}[\mathbb{1}_{xy}] \leq 4/\sqrt{cn}$ for all but at most $n^{1+\delta} < n^{3/2}$ pairs $x, y \in V(G)$. Thus,

$$\mathbb{E}[I] \leq n^{1+\delta} + \frac{4}{\sqrt{cn}} \cdot n^2 \leq \frac{5}{\sqrt{c}} n^{3/2},$$

and so there is a choice for U such that $I \leq 5n^{3/2}/\sqrt{c}$. Now let r be the number of vertices of distinct degrees in H. Let n_k be the number of vertices in H with degree k and note that $\sum_k n_k = m$. By Jensen's inequality

$$5n^{3/2}/\sqrt{c} \geq I = \sum_k \binom{n_k}{2} \geq r\binom{\frac{1}{r}\sum_k n_k}{2} \geq \frac{m^2}{4r} \geq \frac{n^2}{64r}.$$

Rearranging yields the desired result. □

Theorem 2.1 now follows immediately from Lemmas 2.2 and 2.3.

3 Subgraphs of Quadratically Many Sizes

Definition 3.1 ((δ, ε)-rich graph)**.**

Theorem 3.1 (Kwan–Sudakov 2020 [])**.** *Every C-Ramsey graph on n vertices contains induced subgraphs of $\Omega(n^2)$ distinct sizes.*

Proof. □

4 The Erdős–McKay Conjecture

4.1 The Full Erdős–McKay Conjecture and Its Resolution

4.2 Quadratic Polynomials of Gaussians

4.3 The Carbery–Wright Theorem

Theorem 4.1 (Carbery–Wright []).

4.4 Fourier Analysis and Characteristic Functions

4.5 Additive Structure Dichotomy

4.6 The Two-Scale Decomposition

4.7 The Buckets Argument

Theorem 4.2 (Kwan–Sah–Sauermann–Sawhney 2023 []).

5 Conclusing Remarks

References

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A Supplementary Proofs